

HAT-P-32b

R-1001

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_161018 · created 2026-07-30T01:22:04+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457679.8609 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.111 +/- 0.029
Transit depth (Rp/Rs)^2	1.23 +/- 0.64 [%]
Orbital Inclination (inc)	84.01 +/- 1.73
Airmass coefficient 1 (a1)	1.14 +/- 0.016
Airmass coefficient 2 (a2)	-0.098 +/- 0.018
Scatter in the residuals of the lightcurve fit is	1.36 %
Best Comparison Star	None
Optimal Aperture	3.4
Optimal Annulus	7.38
Transit Duration (day)	0.1122 +/- 0.0088

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.111 ± 0.029 vs 0.1489 (deviation 0.81σ)
Duration fit vs published	0.1122 d vs 0.1304 d (deviation 1.28σ)
Mid-time O–C	2.1 min
Depth SNR	2.4
Residual scatter	1.36 %

Light curve

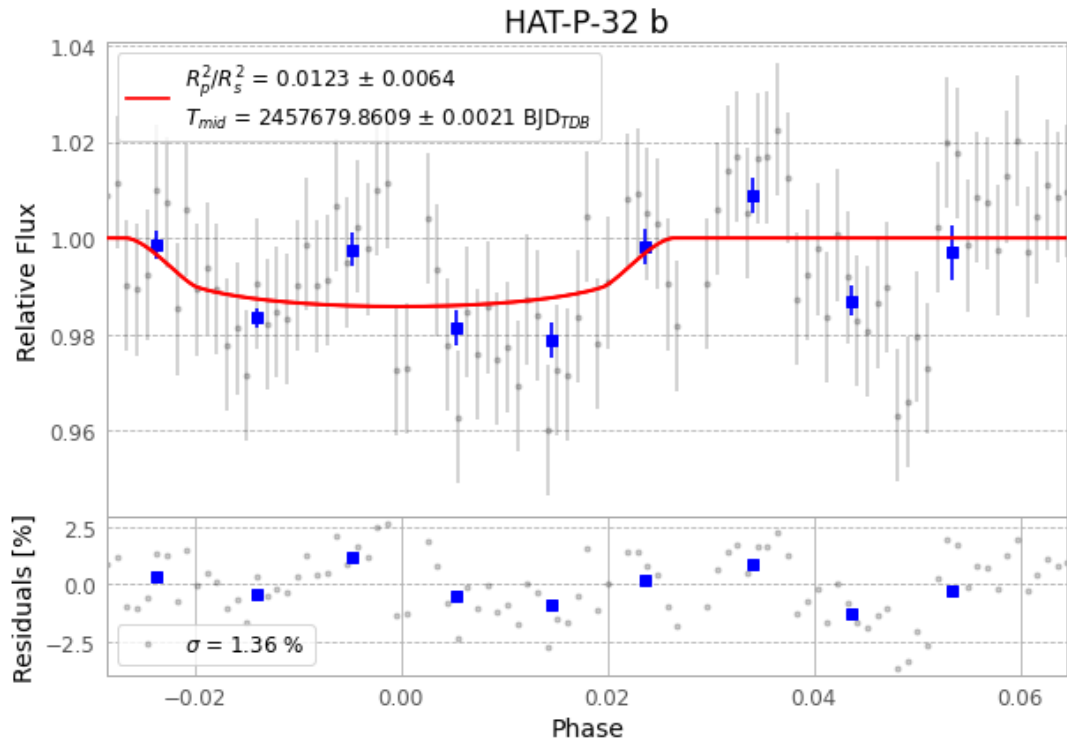


Figure 1 — FinalLightCurve_HAT-P-32 b_2016-10-18.png (SHA-256 c54b47b1c853173f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [186, 270], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ad20ba9baf9707e1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

97 FITS frames (64 MB), HATP-32161018070312.FITS → HATP-32161018115112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-161018.log (369635e47dad...), FinalLightCurve_HAT-P-32 b_2016-10-18.png (c54b47b1c853...), FinalLightCurve_HAT-P-32 b_2016-10-18.csv (cabdd58fe6f4...)

Compute resources

Wall time 80.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 01:22Z.